标题

5. Does the following program represent an algorithm in the strict sense? Why or why not?

```
Count = 0
while (Count != 5):
   Count = Count + 2
```

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Count = 0
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  Count = Count + 2
```

5. No. The process described will never terminate because the value of Count will never be 5.

标题

7. Rewrite the following program segment using a `repeat` structure rather than a `while` structure. Be sure the new version prints the same values as the original.
Initialization: `num = 0`

```
while (num < 50):
   if(num is Odd):
         print(num is Odd)
   num = num + 1
```

标题

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Initialization: `num = 0`

```
while (num < 50):
   if(num is Odd):
        print(num is Odd)
   num = num + 1
```

7. One answer would be

```
Count = 2
repeat:
     print(Count)
      Count = Count + 1
     until (Count >= 7)
```

标题

11. Design an algorithm for finding all the factors of a positive integer. For example, in the case of the integer 12, your algorithm should report the values 1, 2, 3, 4, 6, and 12.

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11. Suppose N is the given integer. Then the following will work. You may want to ask your students how this solution could be made more efficient.

```
X = 1
while (X ≤ N):
    if (X divides N):
        print(X)
    X = X + 1
```

标题

23. What sequence of numbers is printed by the following algorithm if the input value is 1?

```
def CodeWrite (num):
  while (num < 100):
    flag = 0
    i = 2
    while(i < int(num/2)):
      if(num % i == 0):
        flag = 1
      i = i + 1
    if flag == 0:
      print(num)
  num = num + 1
```

标题

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  while (num < 100):
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      if(num % i == 0):
        flag = 1
      i = i + 1
    if flag == 0:
      print(num)
  num = num + 1
```

23. Its output will be 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89.

标题

25. What letters are interrogated by the binary search (Figure 5.14) if it is applied to the list A, B, C, D, E, F, G, H, I, J, K, L, M, N, O when searching for the value J? What about searching for the value Z?

标题

25. What letters are interrogated by the binary search (Figure 5.14) if it is applied to the list A, B, C, D, E, F, G, H, I, J, K, L, M, N, O when searching for the value J? What about searching for the value Z?

25. When searching for J: H, L, J

When searching for Z: H, L, N, 0

标题

34. Call the function `CodeWrite` (defined below) with the value 100 and record the values that are printed.

```
def CodeWrite(N):
  if (N > 1):
    print(N)
    CodeWrite("N / 2")
  print(N + 1)
```

34. Call the function `CodeWrite` (defined below) with the value 100 and record the values that are printed.

```
def CodeWrite(N):
  if (N > 1):
    print(N)
    CodeWrite(“N / 2”)
  print(N + 1)
```

34. 3, 1, 0, 2, 4

35. Call the function `MysteryPrint` (defined below) with the value 2 and record the values that are printed.

```
def MysteryPrint(N):
    if (N > 0):
        print(N)
        MysteryPrint(N – 2)
    else:
        print(N)
        if (N > –1):
            MysteryPrint(N + 1)
```

标题

35. Call the function MysteryPrint (defined below) with the value 2 and record the values that are printed.

```
def MysteryPrint(N):
  if (N > 0):
    print(N)
    MysteryPrint(N – 2)
  else:
    print(N)
    if (N > –1):
      MysteryPrint(N + 1)
```

35. 2, 0, 1, -1

标题

36. Design an algorithm to generate the sequence of positive integers (in increasing order) whose only prime divisors are 2 and 3; that is, your program should produce the sequence 2, 3, 4, 6, 8, 9, 12, 16, 18, 24, 27, ... Does your program represent an algorithm in the strict sense?

36. Design an algorithm to generate the sequence of positive integers (in increasing order) whose only prime divisors are 2 and 3; that is, your program should produce the sequence 2, 3, 4, 6, 8, 9, 12, 16, 18, 24, 27, ... Does your program represent an algorithm in the strict sense?

36.

```
Trial = 2
repeat:
     if (Trial % 2 is 0 or Trial % 3 is 0):
         print(Trial)
     Trial = Trial + 1
    until (2 = 3)
```